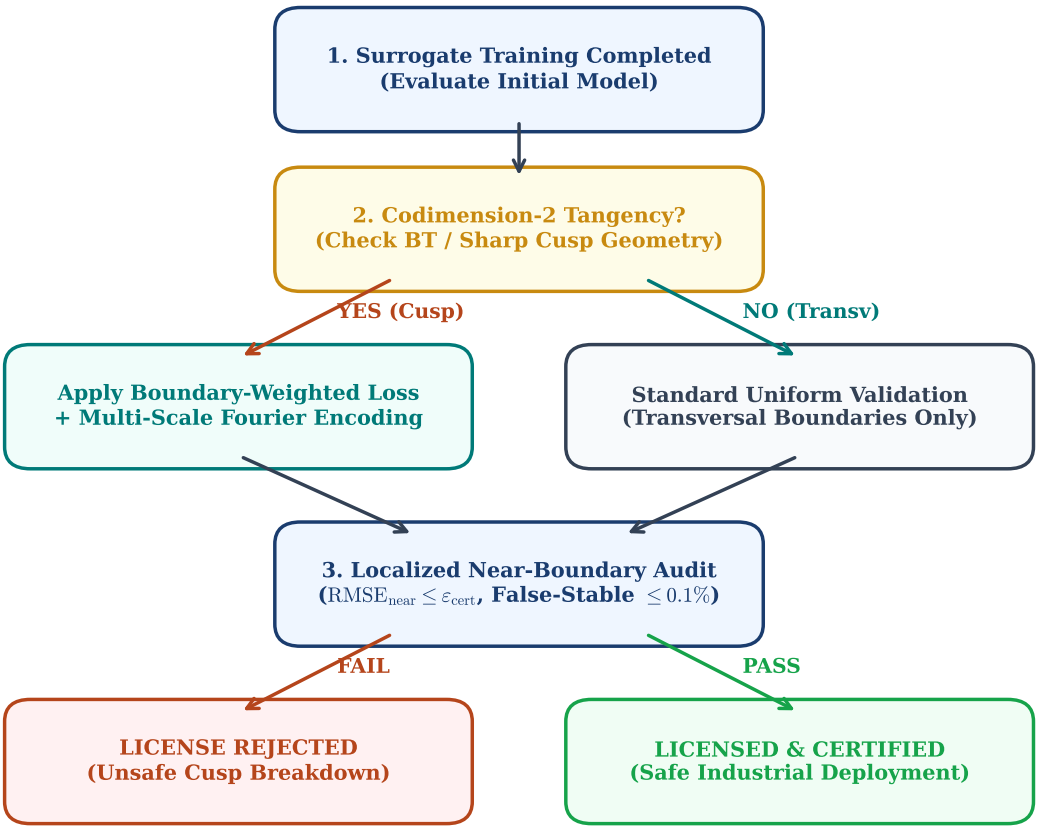


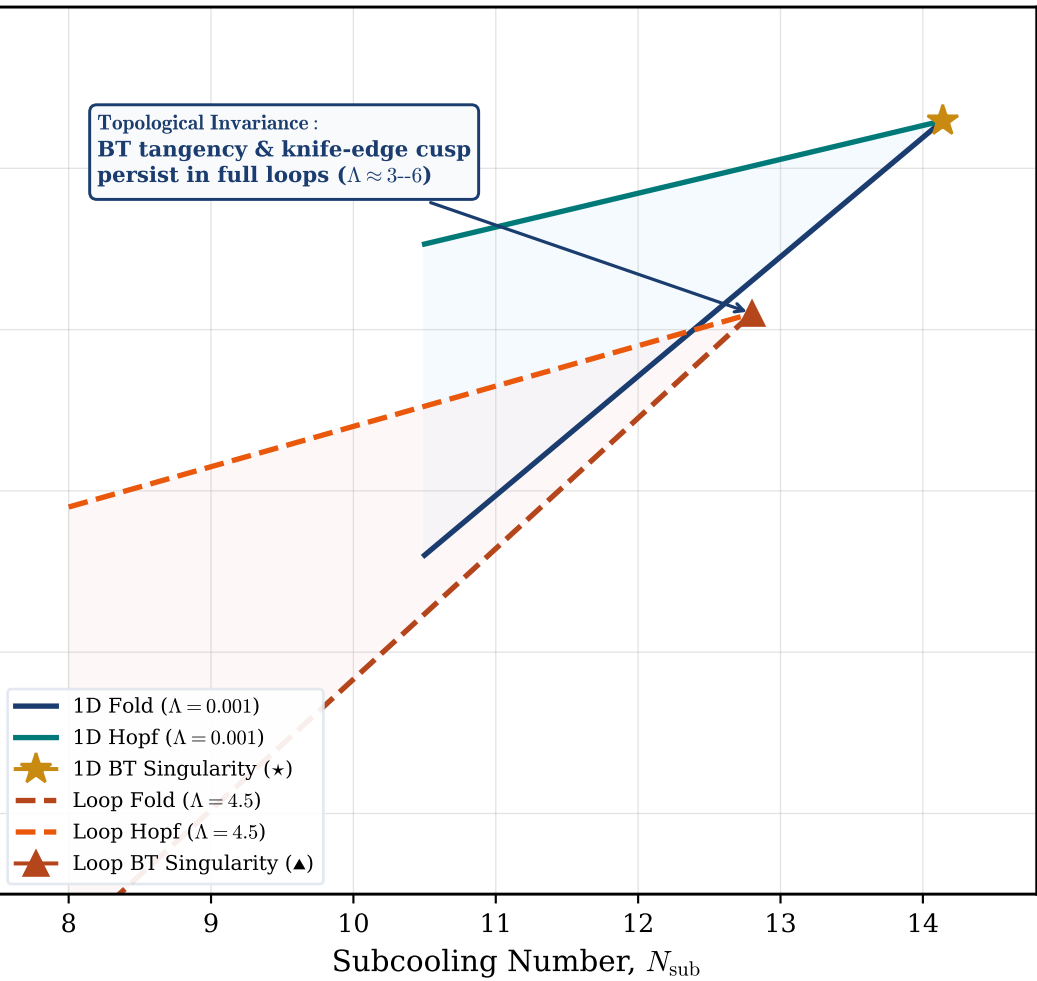
(a) The Four Golden Rules for Safety-Critical Surrogates



(b) Industrial AI Digital Twin Licensing Decision Tree



(c) Scale-Up Invariance: 1D Channel vs. Natural Loop



(d) Cross-Domain Multi-Physics Universality Map

